Measuring Mitosis

Teacher guide | Biology 2e, Chapter 10 | 40-55 minutes

Learning targets and timing

What can a snapshot of dividing cells tell us about the cell cycle?

Load the prepared slide, adjust illumination and focus, and inspect cells at higher power. Classify and collect Field A, then Field B, before exploring microscope controls and cycle-duration assumptions separately. Use stage frequencies to estimate time in each stage under an explicitly assumed cycle duration.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

The student pages open as guided virtual investigations. Teacher customization is deferred to a future teacher workspace. Use this worksheet to assign classroom components separately; discuss data-source differences when comparing results. Existing hybrid links still work, but the student page no longer creates them.

Materials

- Prepared onion root-tip slides, microscope, stage tally sheet

- Alternatively: the app's two schematic fields, with no equipment needed

- Calculator; teacher-selected cycle duration (24 hours is an assumption, not a measured value)

Classroom procedure

1. Review the reference diagrams and stage definitions. Interphase includes G1, S, and G2; these are not distinguished in this activity.

2. Locate a root-tip meristem on a prepared slide at low magnification, then increase magnification to resolve chromosomes. Use fine focus as appropriate.

3. Choose a systematic field and count every interpretable cell without selecting only dividing cells. Record excluded or ambiguous cells in notes.

4. Classify cells as interphase, prophase (including prometaphase), metaphase, anaphase, or telophase. Record a second non-overlapping field.

5. Calculate each stage count / total count x 100. Multiply that fraction by the assumed cycle duration. Explain the assumptions needed to turn a snapshot into a time estimate.

Safety: Use prepared slides only. Carry microscopes with two hands and handle slide edges carefully. The classroom route does not require stains, acids, or cutting root tips.

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Expected reasoning and troubleshooting

The schematic fields together contain 35 interphase, 2 prophase, 1 metaphase, 1 anaphase, and 1 telophase cells (40 total). Order is shuffled per attempt. Review feedback appears only after all cells in a virtual field are classified.

Revised percentages are 87.5% interphase, 5% prophase, and 2.5% each for metaphase, anaphase and telophase (12.5% mitotic index). This is the smallest 40-cell teaching allocation with every mitotic stage represented and prophase most common among dividing cells. Ali et al. (2019) report treatment-dependent mitotic indices of 6.26-11.63% in their experiment; this supports reducing the former 32.5%, but does not validate our exact allocation. Old saved attempts retain their original fields; reset starts revised fields.

At an assumed 24-hour cycle, the modeled durations are 21.0, 1.2, 0.6, 0.6, and 0.6 hours respectively. These are worked examples for the schematic sample, not reference durations for real roots.

Score the quality of morphological evidence and limitations. Classroom counts should not be corrected to match the schematic distribution. In this simplified key, prometaphase is pooled with prophase.

Model boundaries

The virtual fields are original teaching diagrams, not microscope photographs or measured samples. Each contains 20 cells. The revised model makes interphase dominant and allows a rare stage to be absent from one small field. Published onion-root mitotic indices vary with growth conditions, treatment and sampling; these curated counts are not a universal expected ratio or a fitted experimental dataset. A zero count does not mean a stage never occurs.

Estimated stage time = stage count / total count x assumed cycle duration. This assumes representative sampling of an asynchronous, steadily cycling population with comparable cycle lengths. Non-cycling cells, sampling bias, and differential cell loss can invalidate the estimate.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: preparation, investigation, and exploration pages where included, plus companion cutouts. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

Onion-root mitotic and phase indices: Ali et al. (2019): Open source

OpenStax Biology 2e, 10.2: Open source

The investigation materials are original; appended cutouts were supplied by the classroom teacher. Simulated observations are not measurements from the linked sources.

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Guided setup and field collection

Students load a prepared slide, turn on illumination, and focus at the 10x objective (100x total). Once the low-power field is clear, the 40x objective (400x total) becomes available. Focus and illumination affect the view; classification is enabled only with a loaded, illuminated, focused high-power view.

The numbered map preserves the identity of 20 cells. Each classification advances to the next unclassified cell. Selecting a map number lets students revise without double-counting. The higher-power view enlarges a selected cell; it does not introduce a different sample.

After all 20 classifications, Review field classifications shows agreement with the model key. Collect data and continue explicitly saves Field A and advances to Field B. Accept collection even with disagreements; the app preserves student classifications rather than substituting the key. After Field B, analyze the totals or enter free exploration.

Exploration and saved work

Practice microscope settings, classify separate illustrated fields, and vary an assumed 1-48 hour cycle. Changing this assumption changes estimates only. Practice field changes clear that practice tally, not collected data. Exports identify practice counts and duration separately. Reload restores setup, selections, and classifications; existing classroom assignments and saved field data remain supported.

Instructional limits and assessment

These are teaching illustrations, not histology photographs or a physical optics model. Focus and brightness controls use arbitrary scales. Cells do not advance through stages while students count. The model key reflects a curated distribution; real samples need not match it.

Assess chromosome evidence, complete sampling, avoidance of double-counting, uncertainty, and explicit assumptions behind duration estimates. Students can use the stage reference during work. Classroom counts should not be corrected to match the virtual key; discuss sampling differences before pooling. The app does not submit a grade.